

# Vehicle Automation Project — Parts List & Phase Plan

## Phase 1 CAN-Controlled Proof Build + Later Phases (for Budget Sign-Off)

IoT Development Team — Car Automation Internship

### 1. What This Phase Is For

Instead of spending the full proposed budget straight away, Phase 1 is a low-cost proof build. Its job is to demonstrate the main thing we were asked to show — that the CAN module can control a moving vehicle from given inputs — on a small, cheap car we build ourselves.

Phase 1 proves three things, practically and on the ground:

- The car moves autonomously from commands sent over the CAN module (forward, stop, brake, target speed).
- It travels a target distance of about 100 metres (estimated).
- It avoids collisions — when an obstacle is in front while moving, it slows and stops.

Once this works and is shown, Phase 2 (the full higher-budget build with encoder motors for precise distance and braking) can be approved with confidence. This document is Phase 1 only.

#### Why cheap first, not the full budget

The point of Phase 1 is to prove the concept and the CAN control chain works before committing the full amount. Most of the core parts (2 ESP32 boards, 2 CAN modules, the GPS) we already own, so Phase 1 only needs a small set of new low-cost parts. If Phase 1 succeeds, Phase 2 is a justified next step; if something needs rethinking, we find out cheaply.

### 2. How It Works — CAN at the Centre

To show the CAN module is genuinely doing the control (not just one board driving the motors internally), the car uses a two-node CAN bus. This is the core of the demonstration:

- Node 1 — the “brain”: ESP32 + MCP2515 CAN module. It receives commands (from our control dashboard over WebSocket) and sends them out as CAN frames.
- Node 2 — the “motor node”: a second ESP32 + second MCP2515. It reads those CAN frames off the bus and drives the servos accordingly.

The two nodes are joined only by the two CAN wires (CAN-H and CAN-L). This means we can physically point at the CAN bus and show a command crossing it to make the car move — a real, visible demonstration of the CAN module's ability, not a simulation.

We already own two ESP32 boards and two MCP2515 CAN modules, so this two-node setup costs us nothing extra.

### 3. Drive Method — Differential Drive

The car uses differential drive: two driven wheels (one on each side) plus a free castor wheel for balance. Steering is done by driving one side faster than the other — no separate steering mechanism is needed. This is the simplest, most reliable way to build a moving robot car quickly, which is exactly what Phase 1 needs.

#### **Servo note — must be continuous-rotation type**

The drive wheels use servo motors, as specified. Important: these must be CONTINUOUS-ROTATION (360°) servos, not standard positional servos. A standard servo (e.g. a normal SG90/MG996R) only turns 0–180° and holds a position — it cannot spin a wheel to drive the car forward. Continuous-rotation servos spin freely and are controlled by speed/direction, which is what a driving wheel needs. This is why the parts list below specifies the continuous-rotation version.

#### **Alternative considered: Ackermann (car-style) steering**

We also considered Ackermann steering — the real-car layout where the rear wheels drive and the front wheels turn using a steering linkage (one continuous-rotation servo to drive, one standard positional servo to steer). It looks and behaves more like an actual car.

We chose NOT to use it for Phase 1, for honest practical reasons: the steering linkage (tie rods, pivoting front axle) is the most mechanically fiddly part to build and the most likely to eat our build time and risk the demo. Phase 1's goal is to prove the CAN control and autonomy, not to fabricate a working steering rack. Differential drive gives us a reliable moving vehicle fast. Ackermann is documented here as a clear upgrade path for a later phase if a more car-like vehicle is wanted.

### 4. Distance Estimation (~100 m)

Because servos have no built-in encoders, the car needs another way to measure how far it has travelled. We use two methods together:

- Primary — wheel encoder: a slotted disc fitted to a drive wheel, read by an IR slot sensor. Counting the slots as the wheel turns gives distance = wheel rotations × wheel circumference. This is the same principle the expensive encoder motors use, added cheaply to our servo wheels. It is accurate over a short 100 m run.
- Secondary — GPS (NEO-6M, already owned): gives an independent outdoor cross-check of distance. Honest limitation: GPS is only accurate to about 2–5 metres and is weak on slow movement, so it is a rough “we've covered about 100 m” check, not a precise measurement.

Together these give a defensible ~100 m estimate. Precise, to-the-metre distance and closed-loop braking is a Phase 2 job (encoder motors).

## 5. Phase 1 Parts List

Prices are approximate India retail (₹). “Already owned” items are listed for a complete picture but cost nothing. Where a specific low-cost part is available from the local student-electronics vendor, its code is noted; other picks can be sourced from any electronics supplier.

### 5.1 Already owned (reused — ₹0)

| Item               | Qty | Role                                 |
|--------------------|-----|--------------------------------------|
| ESP32 board        | 2   | Node 1 (brain) + Node 2 (motor node) |
| MCP2515 CAN module | 2   | One per node — the CAN bus itself    |
| NEO-6M GPS         | 1   | Secondary distance cross-check       |

### 5.2 Drive (new)

| Item                                                      | Qty   | Key spec                                              | Approx ₹      |
|-----------------------------------------------------------|-------|-------------------------------------------------------|---------------|
| Continuous-rotation servo (360°), high-torque, metal gear | 2     | ~9–11 kg·cm, 4.8–6V, 360° continuous (NOT positional) | ~300 ea (600) |
| Wheel to fit servo horn                                   | 2     | 65–100 mm dia., rubber grip                           | ~40 ea (80)   |
| Castor / free wheel (front balance)                       | 1     | Swivel castor                                         | ~30           |
| Servo mount bracket + horn hardware                       | 1 set | Brackets, screws, servo horns                         | ~80           |

### 5.3 Chassis & mechanical (new)

| Item                           | Qty   | Key spec                                                           | Approx ₹ |
|--------------------------------|-------|--------------------------------------------------------------------|----------|
| Robot car chassis / base plate | 1     | 2-wheel-drive layout, mounting holes, space for 2 boards + battery | ~300–450 |
| M3 standoffs, screws, nuts     | 1 set | For mounting boards and sensors                                    | ~80      |
| Double-sided tape / zip ties   | 1 set | Securing battery and modules                                       | ~50      |

### 5.4 Distance sensing (new)

| Item                                        | Qty | Key spec                                         | Approx ₹     |
|---------------------------------------------|-----|--------------------------------------------------|--------------|
| IR slot speed sensor + encoder disc (LM393) | 2   | One per drive wheel — slotted-disc wheel encoder | ~60 ea (120) |
| GPS (NEO-6M)                                | 1   | Already owned — reused                           | 0            |

### 5.5 Collision avoidance (new)

| Item                                  | Qty | Key spec                                      | Approx ₹ |
|---------------------------------------|-----|-----------------------------------------------|----------|
| HC-SR04 ultrasonic sensor             | 1   | 2–400 cm range, 5V — front obstacle detection | ~60      |
| Ultrasonic mounting holder            | 1   | Front-facing bracket                          | ~30      |
| (Optional upgrade) VL53L0X ToF sensor | 1   | Laser, narrow beam, more precise — optional   | ~450     |

### 5.6 Power (new)

| Item                         | Qty   | Key spec                                                                                     | Approx ₹     |
|------------------------------|-------|----------------------------------------------------------------------------------------------|--------------|
| 18650 Li-ion cell            | 2     | ~7.4V pack (2S), for servos + logic                                                          | ~80 ea (160) |
| 2-cell 18650 holder          | 1     | Holds the pack                                                                               | ~30          |
| Buck converter (LM2596, 3A+) | 1     | Steps battery down to clean 5–6V. Servos and ESP32 run from this — NOT the battery directly. | ~100         |
| On/off switch + inline fuse  | 1 set | Safe power cut-off                                                                           | ~60          |

### 5.7 Wiring & misc (new)

| Item                                     | Qty   | Key spec                              | Approx ₹ |
|------------------------------------------|-------|---------------------------------------|----------|
| Jumper wires (M-M, M-F)                  | 1 set | Connections                           | ~80      |
| Small perfboard / breadboard             | 1     | Mounting the CAN wiring / power rails | ~60      |
| CAN bus wire (twisted pair) + connectors | 1 set | CAN-H / CAN-L between the two nodes   | ~60      |

## 6. Cost Summary

| Section                                | New parts (approx ₹) |
|----------------------------------------|----------------------|
| Drive (servos, wheels, castor, mounts) | ~790                 |
| Chassis & mechanical                   | ~450                 |
| Distance sensing (wheel encoders)      | ~120                 |
| Collision avoidance (ultrasonic)       | ~90                  |
| Power (battery, buck, switch)          | ~350                 |
| Wiring & misc                          | ~200                 |
| <b>Phase 1 total (new parts only)</b>  | <b>≈ ₹2,000</b>      |

| Section                               | New parts (approx ₹) |
|---------------------------------------|----------------------|
| Already owned (2× ESP32, 2× CAN, GPS) | ₹0                   |
| (Optional ToF sensor upgrade)         | + ~450               |

This is well under the full Phase 2 build (roughly ₹14,000), which is the point — prove the concept and the CAN control cheaply first, then scale up once it is demonstrated.

## 7. What Phase 1 Will Demonstrate

- A command sent from our control dashboard travels over the CAN bus (visibly, node-to-node) and makes the car move — proving the CAN module controls the vehicle.
- The car drives forward autonomously, and stops / brakes on command.
- It estimates distance travelled toward ~100 m using the wheel encoder, cross-checked with GPS.
- It detects an obstacle ahead while moving and slows/stops to avoid a collision.

Honest limits of Phase 1 (deferred to Phase 2): distance is an estimate, not to-the-metre precision; braking is basic stop, not measured closed-loop deceleration; and “steer around” obstacle avoidance is a later goal — Phase 1 does detect-and-stop.

## 8. Next Steps

- Get sign-off on this Phase 1 parts list (~₹2,000).
- Order the new parts; assemble the differential-drive car and the two-node CAN wiring.
- Bench-test: CAN command → servo movement, then wheel-encoder distance, then ultrasonic stop.
- Demonstrate the ~100 m autonomous run with collision avoidance.
- On success, propose Phase 2 (encoder motors, precise distance and braking) as the funded next stage.

## PART B — Later Phases (Full Build)

### 9. Later Phases — Overview

Phase 1 above is the low-cost proof build. The sections below list the parts for the full project, to be ordered once Phase 1 has been demonstrated and the larger budget is approved. They are included here so the complete picture — every phase and every part, including the real-car metrics sensing (ELM327 + TPMS) — is in one document.

The later phases are: Phase 2 — the full autonomous car with encoder motors (for precise distance and braking that servos cannot give); Phase 3 & 4 — the autonomy sensors mounted on that car; and the real-car sensing track — reading live metrics from an actual car. Prices are approximate India retail (₹) and will drift  $\pm 10\text{--}15\%$  by order time; treat as budgeting numbers.

### 10. Phase 2 — Full Autonomous Car (Encoder-Motor Build)

| <b>Encoder-motor requirement</b><br>Unlike the Phase 1 servos, Phase 2 uses DC motors with BUILT-IN ENCODERS. This is essential: without encoders the car cannot measure precise speed or distance, so the accurate 100 m run and controlled closed-loop braking are impossible. No plain (non-encoder) gear motors are listed here — both motor options below have genuine built-in quadrature encoders. |     |                                                                                                                        |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------|------------------|
| Item                                                                                                                                                                                                                                                                                                                                                                                                      | Qty | Key spec                                                                                                               | Approx ₹         |
| Chassis — Option A (budget)                                                                                                                                                                                                                                                                                                                                                                               | 1   | 4WD robot car chassis, acrylic/metal double-deck                                                                       | 499–900          |
| Chassis — Option B (rough ground)                                                                                                                                                                                                                                                                                                                                                                         | 1   | Heavier metal 4WD chassis. (True RC-grade chassis not well stocked in India — import lead time; flagged for decision.) | 800–1,500        |
| Encoder motor — Option A                                                                                                                                                                                                                                                                                                                                                                                  | 2   | Cytron SPG30E-60K, 12V 75RPM, quadrature encoder                                                                       | 1,869 ea (3,738) |
| Encoder motor — Option B (recommended)                                                                                                                                                                                                                                                                                                                                                                    | 2   | RMCS-3015, 12V 300RPM 9kg.cm, integrated quad encoder — more torque for a loaded car                                   | 1,500 ea (3,000) |
| Wheels (65mm, matched to shaft)                                                                                                                                                                                                                                                                                                                                                                           | 2   | Silicone grip, nylon hub                                                                                               | 150 ea (300)     |
| Couplers (6mm hex)                                                                                                                                                                                                                                                                                                                                                                                        | 2   | Motor shaft to wheel                                                                                                   | 70 ea (140)      |
| Motor driver — Option A (TB6612FNG)                                                                                                                                                                                                                                                                                                                                                                       | 1   | Light duty; only safe with Motor Option A                                                                              | 200–280          |
| Motor driver — Option B (BTS7960)                                                                                                                                                                                                                                                                                                                                                                         | 2   | 43A H-bridge, one per motor — recommended, esp. with Motor Option B                                                    | 300–400 ea (700) |
| LiPo battery (3S 2200mAh)                                                                                                                                                                                                                                                                                                                                                                                 | 1   | 11.1V, 30C, balance plug                                                                                               | 1,549            |
| Buck converter (LiPo to 5V)                                                                                                                                                                                                                                                                                                                                                                               | 1   | LM2596 5A — powers ESP32, not off LiPo directly                                                                        | ~150             |

|                                  |       |                                                            |             |
|----------------------------------|-------|------------------------------------------------------------|-------------|
| LiPo balance charger             | 1     | IMAX B6-type, 1–6S (copy version)                          | 1,300–1,500 |
| Wiring, connectors, switch, fuse | 1 set | XT60, silicone wire, rocker switch, inline fuse            | 400–500     |
| (Optional) MG996R steering servo | 1     | Standard positional — only if Ackermann steering is chosen | 300–350     |

Phase 2 subtotal (recommended path: Chassis A + Motor B + Driver B): ≈ ₹8,688 (+ ~₹350 if Ackermann steering servo is added). Lock motor + driver + battery together as one decision — do not pair the higher-current Motor Option B with the light TB6612FNG driver.

## 11. Phase 3 & 4 — Autonomy Sensors

These sensors make the Phase 2 car autonomous (obstacle avoidance, straight-line heading). They are mounted on the car; the GPS and CAN module are reused from what we already own.

| Item                                     | Qty   | Key spec                                                                     | Approx ₹ |
|------------------------------------------|-------|------------------------------------------------------------------------------|----------|
| Obstacle sensor — Option A (HC-SR04)     | 1     | Ultrasonic, 2–400cm, cheap and simple                                        | 70–90    |
| Obstacle sensor — Option B (VL53L1X ToF) | 1     | Laser ToF, up to 4m, narrow beam — recommended for collision timing at speed | 550–570  |
| IMU (MPU6050)                            | 1     | Heading hold + hard-brake detection                                          | ~150     |
| GPS (NEO-6M)                             | 1     | Already owned — reused                                                       | 0        |
| CAN module (MCP2515)                     | 1     | Already owned — reused                                                       | 0        |
| Jumper wires / mounting hardware         | 1 set | M3 standoffs, wires                                                          | ~200     |

Phase 3 & 4 subtotal (recommended, VL53L1X): ≈ ₹920 (≈ ₹440 if HC-SR04 is used instead).

## 12. Real-Car Sensing — Vehicle Metrics (ELM327 + TPMS)

This is a separate track that reads live data from an ACTUAL car through its OBD-II port and its tyre sensors — the vehicle-metrics half of the project (RPM, speed, fuel, tyre pressure/temperature, etc.). It plugs into a real car, not our built platform, so it needs a real car available for hands-on testing — schedule that in parallel with ordering.

| Item                                        | Qty | Key spec                                                                                                          | Approx ₹    |
|---------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------|-------------|
| ELM327 OBD-II adapter (USB)                 | 1   | FTDI USB, all OBD2 protocols — USB for stable wired link                                                          | 1,800–2,000 |
| OBD-II extension cable                      | 1   | 16-pin male-to-female, ~1–1.5m                                                                                    | 400–500     |
| TPMS sensors (external valve-cap, set of 4) | 4   | 433MHz broadcast. Sold as kits with a display we don't need. Packet format varies by brand — confirm brand before | 2,000–2,500 |

| Item                      | Qty | Key spec                                                                    | Approx ₹ |
|---------------------------|-----|-----------------------------------------------------------------------------|----------|
|                           |     | ordering (Person E's decoding depends on it).                               |          |
| 433MHz RF receiver module | 1   | Superheterodyne (SRX882) — picks up TPMS broadcasts for the ESP32 to decode | 100–150  |

Real-car sensing subtotal: ≈ ₹4,725 (using range midpoints).

### 13. Overall Budget Summary (All Phases)

| Phase                                              | New parts (approx ₹) |
|----------------------------------------------------|----------------------|
| Phase 1 — CAN-controlled proof build (servos)      | ~2,000               |
| Phase 2 — Full autonomous car (encoder motors)     | ~8,688               |
| Phase 3 & 4 — Autonomy sensors                     | ~920                 |
| Real-car sensing — ELM327 + TPMS                   | ~4,725               |
| <b>Grand total, all phases (recommended paths)</b> | <b>≈ ₹16,300</b>     |
| Already owned (2× ESP32, 2× CAN, GPS)              | ₹0                   |

Recommended sequencing: approve and run Phase 1 (~₹2,000) first to prove the CAN control and autonomy cheaply. On success, approve the later phases. Phase 1 and Phase 2 both include a chassis and an obstacle sensor because they are two separate builds (cheap proof vs. full car); the Phase 1 ultrasonic can carry over to save a little on the later phase.